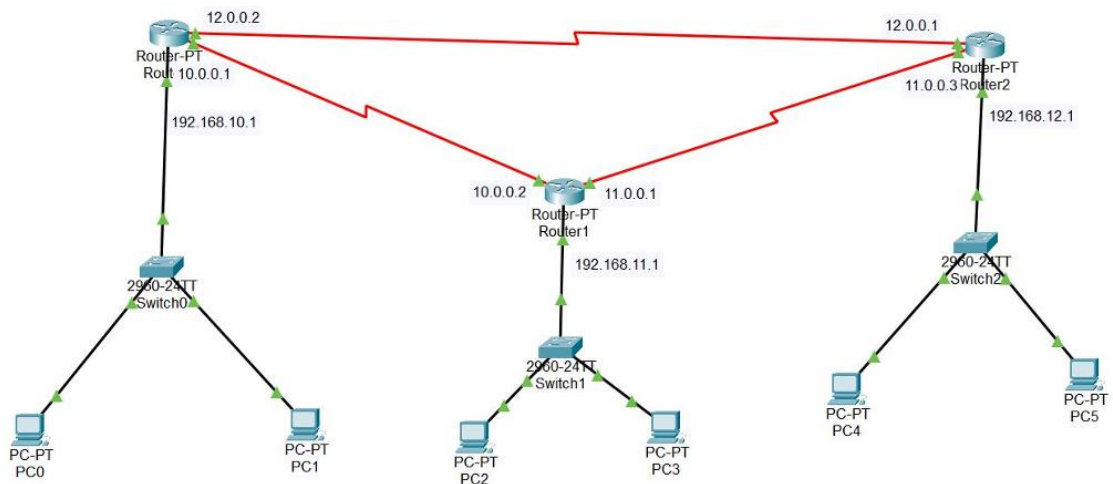




Dynamic routing using RIP

Based on your Packet Tracer topology, you have **3 routers connected in a triangle** and each router has one LAN.

Network Topology

Router LAN Interface WAN Interface 1 WAN Interface 2

Router0	192.168.10.1/24	10.0.0.1/30	12.0.0.2/30
Router1	192.168.11.1/24	10.0.0.2/30	11.0.0.1/30
Router2	192.168.12.1/24	11.0.0.2/30	12.0.0.1/30

Step 1: Build the Topology

Connect devices as shown.

- 3 Routers
- 3 Switches
- 6 PCs

Step 2: Configure IP Address on PCs

LAN 1

PC0

IP Address : 192.168.10.2
Subnet Mask: 255.255.255.0
Gateway : 192.168.10.1

PC1

IP Address : 192.168.10.3
Subnet Mask: 255.255.255.0
Gateway : 192.168.10.1

LAN 2

PC2

IP Address : 192.168.11.2
Subnet Mask: 255.255.255.0
Gateway : 192.168.11.1

PC3

IP Address : 192.168.11.3
Subnet Mask: 255.255.255.0
Gateway : 192.168.11.1

LAN 3

PC4

IP Address : 192.168.12.2
Subnet Mask: 255.255.255.0

Gateway : 192.168.12.1
PC5
IP Address : 192.168.12.3
Subnet Mask: 255.255.255.0
Gateway : 192.168.12.1

Step 3: Configure Router0

```
enable
configure terminal
LAN Interface
interface fa0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

Serial to Router1

```
interface s0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit
```

(Configure clock rate only if this side has the DCE cable.)

Serial to Router2

```
interface s0/0/1
ip address 12.0.0.2 255.255.255.252
clock rate 64000
no shutdown
exit
```

(Again, use clock rate only on the DCE side.)

Step 4: Configure Router1

```
enable
configure terminal
LAN
interface fa0/0
ip address 192.168.11.1 255.255.255.0
no shutdown
exit
```

Serial to Router0

```
interface s0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
```

Serial to Router2

```
interface s0/0/1
ip address 11.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit
```

(Only if DCE.)

Step 5: Configure Router2

```
enable
configure terminal
LAN
interface fa0/0
ip address 192.168.12.1 255.255.255.0
no shutdown
```

exit

Serial to Router1

interface s0/0/0

ip address 11.0.0.2 255.255.255.252

no shutdown

exit

Serial to Router0

interface s0/0/1

ip address 12.0.0.1 255.255.255.252

no shutdown

exit

Step 6: Verify Interfaces

On each router:

show ip interface brief

Expected:

Interface	IP Address	Status
Fa0/0	Up	Up
S0/0/0	Up	Up
S0/0/1	Up	Up

Step 7: Configure RIP Version 2

Router0

enable

configure terminal

router rip

version 2

no auto-summary

network 192.168.10.0

network 10.0.0.0

network 12.0.0.0

end

write

Router1

enable

configure terminal

router rip

version 2

no auto-summary

network 192.168.11.0

network 10.0.0.0

network 11.0.0.0

end

write

Router2

enable

configure terminal

router rip

```
version 2
no auto-summary
```

```
network 192.168.12.0
network 11.0.0.0
network 12.0.0.0
```

```
end
write
```

Step 8: Verify RIP

```
show ip protocols
```

You should see:

Routing Protocol is "rip"

Version 2

Automatic network summarization is not in effect

Routing for Networks:

192.168.10.0

10.0.0.0

12.0.0.0

Step 9: Check Routing Table

```
show ip route
```

Router0 should learn:

R 192.168.11.0

R 192.168.12.0

Router1 should learn:

R 192.168.10.0

R 192.168.12.0

Router2 should learn:

R 192.168.10.0

R 192.168.11.0

R indicates the routes were learned through RIP.

Step 10: Test Connectivity

From **PC0**:

```
ping 192.168.11.2
```

Expected:

Reply from 192.168.11.2

Then:

```
ping 192.168.12.2
```

Expected:

Reply from 192.168.12.2

You can also test from **PC5**:

```
ping 192.168.10.2
```

Step 11: Check RIP Routes

On any router:

```
show ip route rip
```

Example output:

R 192.168.11.0

R 192.168.12.0

Step 12: Save the Configuration

copy running-config startup-config
or
write memory

Final Network Summary

Device IP Address

Router0 LAN 192.168.10.1

Router1 LAN 192.168.11.1

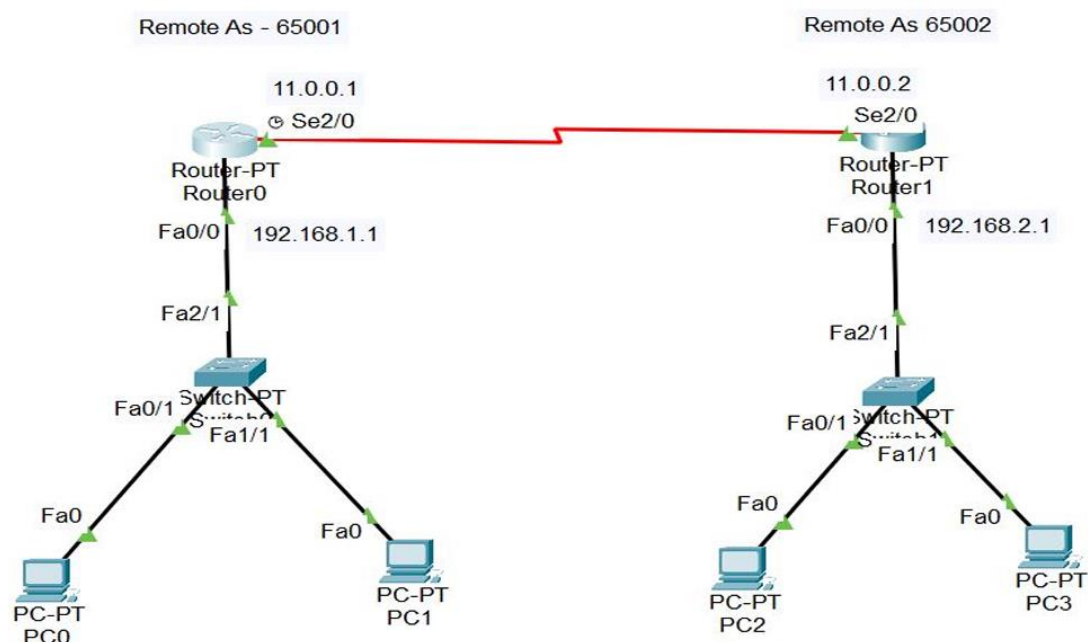
Router2 LAN 192.168.12.1

R0 ↔ R1 10.0.0.1 ↔ 10.0.0.2

R1 ↔ R2 11.0.0.1 ↔ 11.0.0.2

R0 ↔ R2 12.0.0.2 ↔ 12.0.0.1

After completing these steps, all PCs in the three LANs should be able to communicate with one another through **RIP Version 2 dynamic routing**.



Configuring Border Gateway Protocol (BGP) for Inter-AS Routing

Below is a **step-by-step procedure to configure BGP (Border Gateway Protocol) for Inter-AS Routing** based on your Packet Tracer topology.

Network Topology

Device	Interface	IP Address	Description
--------	-----------	------------	-------------

Router0	Fa0/0	192.168.1.1/24	LAN
---------	-------	----------------	-----

Router0	Se2/0	11.0.0.1/30	WAN
---------	-------	-------------	-----

Router1	Se2/0	11.0.0.2/30	WAN
---------	-------	-------------	-----

Router1	Fa0/0	192.168.2.1/24	LAN
---------	-------	----------------	-----

Autonomous Systems (AS)

- Router0 → AS 65001
- Router1 → AS 65002

Step 1: Configure PCs

PC0

IP Address : 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

PC1

IP Address : 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

PC2

IP Address : 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

PC3

IP Address : 192.168.2.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Step 2: Configure Router0 (AS 65001)

enable

! Enter privileged EXEC mode.

configure terminal

! Enter global configuration mode.

interface fastethernet0/0

! Select LAN interface.

ip address 192.168.1.1 255.255.255.0

! Assign IP address to LAN.

no shutdown

! Enable the interface.

exit

interface serial2/0

! Select serial interface connected to Router1.

ip address 11.0.0.1 255.255.255.252

! Assign WAN IP address.

clock rate 64000

! Set clock rate only if this router has the DCE cable.

no shutdown

! Enable the interface.

exit

Step 3: Configure Router1 (AS 65002)

enable

! Enter privileged EXEC mode.

configure terminal

! Enter global configuration mode.

interface fastethernet0/0

! Select LAN interface.

ip address 192.168.2.1 255.255.255.0

! Assign LAN IP address.

no shutdown

! Enable the interface.

exit

interface serial2/0

! Select serial interface connected to Router0.

ip address 11.0.0.2 255.255.255.252

! Assign WAN IP address.

no shutdown

! Enable the interface.

exit

Step 4: Configure BGP on Router0

router bgp 65001

! Start BGP process with AS Number 65001.

neighbor 11.0.0.2 remote-as 65002

! Define Router1 as the BGP neighbor in AS 65002.

network 192.168.1.0 mask 255.255.255.0

! Advertise the local LAN network into BGP.

exit

Step 5: Configure BGP on Router1

router bgp 65002

! Start BGP process with AS Number 65002.

neighbor 11.0.0.1 remote-as 65001

! Define Router0 as the BGP neighbor in AS 65001.

network 192.168.2.0 mask 255.255.255.0

! Advertise the local LAN network into BGP.

exit

Step 6: Save the Configuration

copy running-config startup-config

! Save the running configuration into NVRAM.

Step 7: Verify BGP Neighbor

On Router0

show ip bgp summary

! Displays the BGP neighbor status and session information.

Expected output:

Neighbor	AS	State/PfxRcd
11.0.0.2	65002	1

On Router1

show ip bgp summary

! Displays the BGP neighbor status.

Expected output:

Neighbor	AS	State/PfxRcd
11.0.0.1	65001	1

If the **State/PfxRcd** column shows a **number** (e.g., **1**) instead of **Idle**, **Active**, or **Connect**, the BGP session has been established successfully.

Step 8: Verify BGP Routes

show ip bgp

! Displays all routes learned and advertised through BGP.

Step 9: Verify Routing Table

show ip route

! Displays the router's routing table.

Expected:

On **Router0**

B 192.168.2.0/24

On **Router1**

B 192.168.1.0/24

B indicates the route was learned through **BGP**.

Step 10: Test Connectivity

From **PC0**

ping 192.168.2.2

! Test connectivity to PC2.

ping 192.168.2.3

! Test connectivity to PC3.

From **PC2**

ping 192.168.1.2

! Test connectivity to PC0.

ping 192.168.1.3

! Test connectivity to PC1.

Successful replies confirm that inter-AS routing via BGP is working.

Step 11: Useful Verification Commands

show ip bgp

! Displays the BGP routing table.

show ip bgp summary

! Displays neighbor status and received prefixes.

show ip route

! Displays all routing table entries.

show running-config

! Displays the active router configuration.

show ip interface brief

! Displays interfaces, IP addresses, and their status.

show ip protocols

! Displays configured routing protocols.

ping 11.0.0.2

! Tests connectivity to the BGP neighbor.

tracert 192.168.2.2

! Displays the path packets take to reach the destination.

copy running-config startup-config
! Saves the current configuration.

Final Network Summary

Router AS Number Advertised Network Neighbor Remote AS

Router0	65001	192.168.1.0/24	11.0.0.2	65002
Router1	65002	192.168.2.0/24	11.0.0.1	65001

Expected Result

-  Router0 and Router1 establish a BGP peering session.
-  Router0 learns **192.168.2.0/24** through BGP.
-  Router1 learns **192.168.1.0/24** through BGP.
-  show ip route displays routes marked with **B** (BGP).
-  All PCs on both LANs can successfully communicate with each other using BGP-based inter-AS routing.